# Lecture 2: Electrical Quantities

In the previous class,

Charge, $q$

$W = Vq$

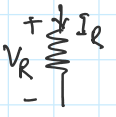
$i = \frac{dq}{dt}$

$p = vi$

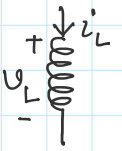
$E_n = \int v\,i\,dt$

Resistance:

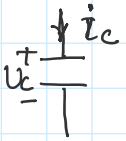
$V = IR$ or $\frac{V}{I} = R$ (Ohms)




Inductance, $L = \frac{v_L}{\frac{di_L}{dt}}$ (Henry)




Capacitance, $C = \frac{\int i\,dt}{v_C}$ [F]



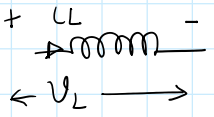

Resistance of a wire: $R = \rho \frac{l}{A}$, $\rho$ = Specific resistivity Ohm-m

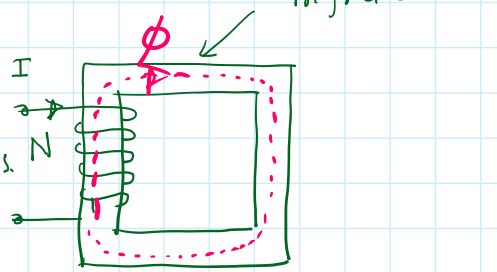
$l$ = length of wire

$A$ = Area of cross section

Inductance $L\frac{di_L}{dt} = v_L$

$L = \frac{v_L}{\frac{di_L}{dt}}$




$\mathcal{F}$ = Magnetomotive force = No of Ampere Turns.

$e$ = Electromotive force

$$e = -N\frac{d\phi}{dt} = -L\frac{di}{dt}$$

$$N\frac{d\phi}{di} = L\frac{di}{dt} \Rightarrow N\phi = Li \Rightarrow L = \frac{N\phi}{i} = \frac{N \cdot B \cdot A}{i} \quad (*)$$

## Ampere's Circuital Law

No of Ampere turns enclosed equals to the closed line integration of magnetic field intensity $\vec{H}$

$$NI = \oint \vec{H} \cdot d\vec{l}$$

$$= H \cdot l$$

$$\boxed{I = \frac{Bl}{N\mu}} \quad (*) \Rightarrow L = \frac{N \cdot B\,A}{Bl/N\mu} = \frac{N^2}{\left(\frac{l}{\mu A}\right)}$$

$\vec{B} = \mu\vec{H}$

$$L = \frac{N^2}{(l/\mu A)} = \frac{N^2}{\mathcal{R}l}$$

$\mathcal{R}l$ = Reluctance of Magnetic circuit $= \frac{l}{\mu A}$

$$\Rightarrow \boxed{L * \mathcal{R}l = N^2}$$

$V = IR$ Electrical ckt

$\phi = \frac{\mathcal{F}}{\mathcal{R}l}$ Magnetic ckt

$\Rightarrow \boxed{L * \mathcal{R}l = N^2}$

$$\boxed{L = \frac{N^2}{\ell/\mu A}}$$

$\gamma = \mathcal{I}\mathcal{R}$ Electrical ckt

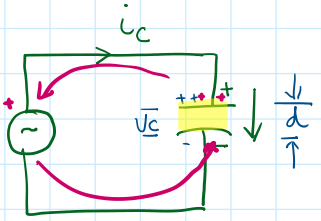
$\mathcal{F} = \phi * \mathcal{R}l$ Magnetic ckt

## Capacitance

$$V_c = \frac{1}{C}\int i_c\,dt$$

$$i_c = C\frac{dV_c}{dt}$$

$$C = \frac{\int i_c\,dt}{V_c} = \frac{q_c}{V_c}$$




$$E = \frac{V_c}{d} = \frac{V}{d}$$

Gauss's Law $\quad q_c = \oint \vec{D}_1 \cdot d\vec{s}$

$$q_c = \epsilon E A$$

$$q_c = \epsilon \cdot \frac{V_c}{d} A$$

$$\boxed{\frac{q_c}{V_c} = \frac{\epsilon A}{d} = C}$$

$$p_c = V_c i_c = V_c \cdot C\frac{dV_c}{dt} = C V_c \frac{dV_c}{dt}$$

$$E_c = \int p_c\,dt = \int C V_c \frac{dV_c}{dt}\,dt = \int C V_c\,dV_c = \frac{1}{2} C V_c^2$$

$$p_L = V_L i_L = L\frac{di_L}{dt} i_L = L i_L \frac{di_L}{dt}$$

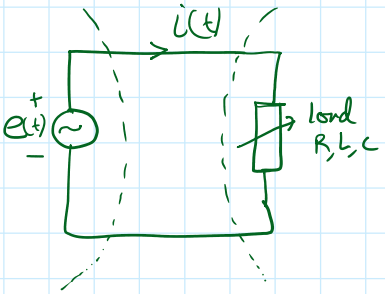
$$E_L = \int p_L\,dt = \frac{1}{2} L i_L^2$$

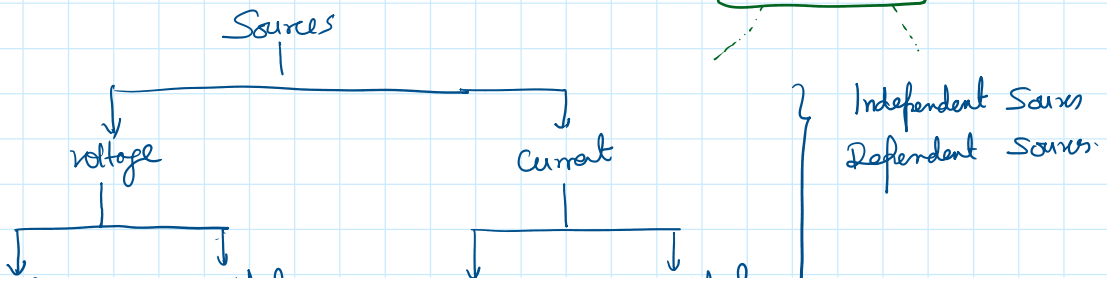
L, C are known as energy storing elements

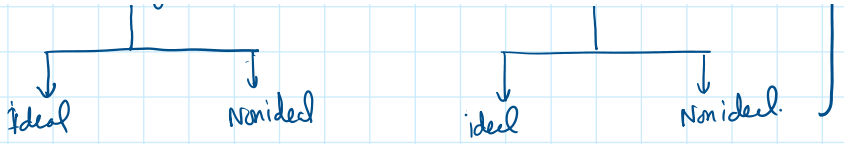
$$p_R = V_R i_R = i_R^2 * R = \frac{V_R^2}{R}$$

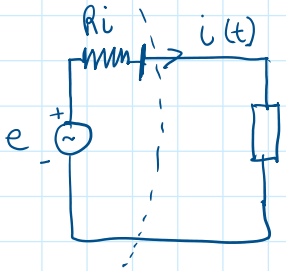
$$E_R = \int p_R\,dt = i_R^2 R t = \frac{V_R^2}{R} t$$

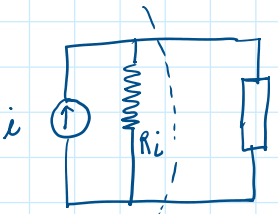
## Sources






$R_i \to \infty$ for ideal current source

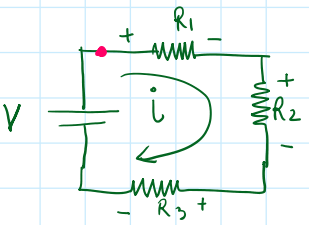
**Dependent Sources:**

| | |
|---|---|
| $V_x = f(V_y)$ | Voltage dependent voltage source. |
| $I_x = f(V_y)$ | Voltage dependent Current source. |
| $V_x = f(I_y)$ | Current dependent voltage source. |
| $I_x = f(I_y)$ | Current dependent current source. |

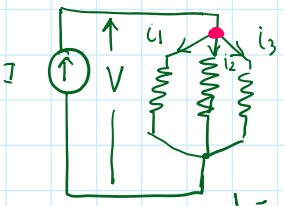
## DC Circuit Analysis

### Kirchhoff's Laws:

1. Kirchhoff's Voltage Law (KVL)
2. Kirchhoff's Current Law (KCL)




A mesh system




A nodal system

KVL: $\sum_{j=1}^{n} V_j = 0$

KCL: $\sum_{j=1}^{n} I_j = 0$